

Informatica® RulePoint
6.1.2

# Getting Started Guide

Informatica RulePoint Getting Started Guide
6.1.2
April 2015

© Copyright Informatica LLC 1998, 2018

This software and documentation contain proprietary information of Informatica Corporation and are provided under a license agreement containing restrictions on use and disclosure and are also protected by copyright law. Reverse engineering of the software is prohibited. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica Corporation. This Software may be protected by U.S. and/or international Patents and other Patents Pending.

Use, duplication, or disclosure of the Software by the U.S. Government is subject to the restrictions set forth in the applicable software license agreement and as provided in DFARS 227.7202-1(a) and 227.7702-3(a) (1995), DFARS 252.227-7013©(1)(ii) (OCT 1988), FAR 12.212(a) (1995), FAR 52.227-19, or FAR 52.227-14 (ALT III), as applicable.

The information in this product or documentation is subject to change without notice. If you find any problems in this product or documentation, please report them to us in writing.

Informatica, Informatica Platform, Informatica Data Services, PowerCenter, PowerCenterRT, PowerCenter Connect, PowerCenter Data Analyzer, PowerExchange, PowerMart, Metadata Manager, Informatica Data Quality, Informatica Data Explorer, Informatica B2B Data Transformation, Informatica B2B Data Exchange Informatica On Demand, Informatica Identity Resolution, Informatica Application Information Lifecycle Management, Informatica Complex Event Processing, Ultra Messaging and Informatica Master Data Management are trademarks or registered trademarks of Informatica Corporation in the United States and in jurisdictions throughout the world. All other company and product names may be trade names or trademarks of their respective owners.

Portions of this software and/or documentation are subject to copyright held by third parties, including without limitation: Copyright DataDirect Technologies. All rights reserved. Copyright © Sun Microsystems. All rights reserved. Copyright © RSA Security Inc. All Rights Reserved. Copyright © Ordinal Technology Corp. All rights reserved.Copyright © Aandacht c.v. All rights reserved. Copyright Genivia, Inc. All rights reserved. Copyright Isomorphic Software. All rights reserved. Copyright © Meta Integration Technology, Inc. All rights reserved. Copyright © Intalio. All rights reserved. Copyright © Oracle. All rights reserved. Copyright © Adobe Systems Incorporated. All rights reserved. Copyright © DataArt, Inc. All rights reserved. Copyright © ComponentSource. All rights reserved. Copyright © Microsoft Corporation. All rights reserved. Copyright © Rogue Wave Software, Inc. All rights reserved. Copyright © Teradata Corporation. All rights reserved. Copyright © Yahoo! Inc. All rights reserved. Copyright © Glyph & Cog, LLC. All rights reserved. Copyright © Thinkmap, Inc. All rights reserved. Copyright © Clearpace Software Limited. All rights reserved. Copyright © Information Builders, Inc. All rights reserved. Copyright © OSS Nokalva, Inc. All rights reserved. Copyright Edifecs, Inc. All rights reserved. Copyright Cleo Communications, Inc. All rights reserved. Copyright © International Organization for Standardization 1986. All rights reserved. Copyright © ej-technologies GmbH. All rights reserved. Copyright © Jaspersoft Corporation. All rights reserved. Copyright © International Business Machines Corporation. All rights reserved. Copyright © yWorks GmbH. All rights reserved. Copyright © Lucent Technologies. All rights reserved. Copyright (c) University of Toronto. All rights reserved. Copyright © Daniel Veillard. All rights reserved. Copyright © Unicode, Inc. Copyright IBM Corp. All rights reserved. Copyright © MicroQuill Software Publishing, Inc. All rights reserved. Copyright © PassMark Software Pty Ltd. All rights reserved. Copyright © LogiXML, Inc. All rights reserved. Copyright © 2003-2010 Lorenzi Davide, All rights reserved. Copyright © Red Hat, Inc. All rights reserved. Copyright © The Board of Trustees of the Leland Stanford Junior University. All rights reserved. Copyright © EMC Corporation. All rights reserved. Copyright © Flexera Software. All rights reserved. Copyright © Jinfonet Software. All rights reserved. Copyright © Apple Inc. All rights reserved. Copyright © Telerik Inc. All rights reserved. Copyright © BEA Systems. All rights reserved. Copyright © PDFlib GmbH. All rights reserved. Copyright © Orientation in Objects GmbH. All rights reserved. Copyright © Tanuki Software, Ltd. All rights reserved. Copyright © Ricebridge. All rights reserved. Copyright © Sencha, Inc. All rights reserved. Copyright © Scalable Systems, Inc. All rights reserved. Copyright © jQWidgets. All rights reserved. Copyright © Tableau Software, Inc. All rights reserved. Copyright© MaxMind, Inc. All Rights Reserved. Copyright © TMate Software s.r.o. All rights reserved.

This product includes software developed by the Apache Software Foundation (http://www.apache.org/), and/or other software which is licensed under various versions of the Apache License (the "License"). You may obtain a copy of these Licenses at http://www.apache.org/licenses/. Unless required by applicable law or agreed to in writing, software distributed under these Licenses is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the Licenses for the specific language governing permissions and limitations under the Licenses.

This product includes software which was developed by Mozilla (http://www.mozilla.org/), software copyright The JBoss Group, LLC, all rights reserved; software copyright © 1999-2006 by Bruno Lowagie and Paulo Soares and other software which is licensed under various versions of the GNU Lesser General Public License Agreement, which may be found at http:// www.gnu.org/licenses/lgpl.html. The materials are provided free of charge by Informatica, "as-is", without warranty of any kind, either express or implied, including but not limited to the implied warranties of merchantability and fitness for a particular purpose.

The product includes ACE(TM) and TAO(TM) software copyrighted by Douglas C. Schmidt and his research group at Washington University, University of California, Irvine, and Vanderbilt University, Copyright (©) 1993-2006, all rights reserved.

This product includes software developed by the OpenSSL Project for use in the OpenSSL Toolkit (copyright The OpenSSL Project. All Rights Reserved) and redistribution of this software is subject to terms available at http://www.openssl.org and http://www.openssl.org/source/license.html.

This product includes Curl software which is Copyright 1996-2013, Daniel Stenberg, <daniel@haxx.se>. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://curl.haxx.se/docs/copyright.html. Permission to use, copy, modify, and distribute this software for any purpose with or without fee is hereby granted, provided that the above copyright notice and this permission notice appear in all copies.

The product includes software copyright 2001-2005 (©) MetaStuff, Ltd. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://www.dom4j.org/ license.html.

The product includes software copyright © 2004-2007, The Dojo Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://dojotoolkit.org/license.

This product includes ICU software which is copyright International Business Machines Corporation and others. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://source.icu-project.org/repos/icu/icu/trunk/license.html.

This product includes software copyright © 1996-2006 Per Bothner. All rights reserved. Your right to use such materials is set forth in the license which may be found at http:// www.gnu.org/software/ kawa/Software-License.html.

This product includes OSSP UUID software which is Copyright © 2002 Ralf S. Engelschall, Copyright © 2002 The OSSP Project Copyright © 2002 Cable & Wireless Deutschland. Permissions and limitations regarding this software are subject to terms available at http://www.opensource.org/licenses/mit-license.php.

This product includes software developed by Boost (http://www.boost.org/) or under the Boost software license. Permissions and limitations regarding this software are subject to terms available at http:/ /www.boost.org/LICENSE_1_0.txt.

This product includes software copyright © 1997-2007 University of Cambridge. Permissions and limitations regarding this software are subject to terms available at http:// www.pcre.org/license.txt.

This product includes software copyright © 2007 The Eclipse Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http:// www.eclipse.org/org/documents/epl-v10.php and at http://www.eclipse.org/org/documents/edl-v10.php.

This product includes software licensed under the terms at http://www.tcl.tk/software/tcltk/license.html, http://www.bosrup.com/web/overlib/?License, http:// www.stlport.org/doc/ license.html, http://asm.ow2.org/license.html, http://www.cryptix.org/LICENSE.TXT, http://hsqldb.org/web/hsqlLicense.html, http://

httpunit.sourceforge.net/doc/ license.html, http://jung.sourceforge.net/license.txt , http://www.gzip.org/zlib/zlib_license.html, http://www.openldap.org/software/release/license.html, http://www.libssh2.org, http://slf4j.org/license.html, http://www.sente.ch/software/OpenSourceLicense.html, http://fusesource.com/downloads/license-agreements/fuse-message-broker-v-5-3- license-agreement; http://antlr.org/license.html; http://aopalliance.sourceforge.net/; http://www.bouncycastle.org/licence.html; http://www.jgraph.com/jgraphdownload.html; http://www.jcraft.com/jsch/LICENSE.txt; http://jotm.objectweb.org/bsd_license.html; . http://www.w3.org/Consortium/Legal/2002/copyright-software-20021231; http://www.slf4j.org/license.html; http://nanoxml.sourceforge.net/orig/copyright.html; http://www.json.org/license.html; http://forge.ow2.org/projects/javaservice/, http://www.postgresql.org/about/licence.html, http://www.sqlite.org/copyright.html, http://www.tcl.tk/software/tcltk/license.html, http://www.jaxen.org/faq.html, http://www.jdom.org/docs/faq.html, http://www.slf4j.org/license.html; http://www.iodbc.org/dataspace/iodbc/wiki/iODBC/License; http://www.keplerproject.org/md5/license.html; http://www.toedter.com/en/jcalendar/license.html; http://www.edankert.com/bounce/index.html; http://www.net-snmp.org/about/license.html; http://www.openmdx.org/#FAQ; http://www.php.net/license/3_01.txt; http://srp.stanford.edu/license.txt; http://www.schneier.com/blowfish.html; http://www.jmock.org/license.html; http://xsom.java.net; http://benalman.com/about/license/; https://github.com/CreateJS/EaselJS/blob/master/src/easeljs/display/Bitmap.js; http://www.h2database.com/html/license.html#summary; http://jsoncpp.sourceforge.net/LICENSE; http://jdbc.postgresql.org/license.html; http://protobuf.googlecode.com/svn/trunk/src/google/protobuf/descriptor.proto; https://github.com/rantav/hector/blob/master/LICENSE; http://web.mit.edu/Kerberos/krb5-current/doc/mitK5license.html; http://jibx.sourceforge.net/jibx-license.html; https://github.com/lyokato/libgeohash/blob/master/LICENSE; https://github.com/hjiang/jsonxx/blob/master/LICENSE; https://code.google.com/p/lz4/; https://github.com/jedisct1/libsodium/blob/master/LICENSE; http://one-jar.sourceforge.net/index.php?page=documents&file=license; https://github.com/EsotericSoftware/kryo/blob/master/license.txt; http://www.scala-lang.org/license.html; https://github.com/tinkerpop/blueprints/blob/master/LICENSE.txt; and http://gee.cs.oswego.edu/dl/classes/EDU/oswego/cs/dl/util/concurrent/intro.html.

This product includes software licensed under the Academic Free License (http://www.opensource.org/licenses/afl-3.0.php), the Common Development and Distribution License (http://www.opensource.org/licenses/cddl1.php) the Common Public License (http://www.opensource.org/licenses/cpl1.0.php), the Sun Binary Code License Agreement Supplemental License Terms, the BSD License (http:// www.opensource.org/licenses/bsd-license.php), the new BSD License (http://opensource.org/licenses/BSD-3-Clause), the MIT License (http://www.opensource.org/licenses/mit-license.php), the Artistic License (http://www.opensource.org/licenses/artistic-license-1.0) and the Initial Developer’s Public License Version 1.0 (http://www.firebirdsql.org/en/initial-developer-s-public-license-version-1-0/).

This product includes software copyright © 2003-2006 Joe Walnes, 2006-2007 XStream Committers. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://xstream.codehaus.org/license.html. This product includes software developed by the Indiana University Extreme! Lab. For further information please visit http://www.extreme.indiana.edu/.

This product includes software Copyright (c) 2013 Frank Balluffi and Markus Moeller. All rights reserved. Permissions and limitations regarding this software are subject to terms of the MIT license.

This Software is protected by U.S. Patent Numbers 5,794,246; 6,014,670; 6,016,501; 6,029,178; 6,032,158; 6,035,307; 6,044,374; 6,092,086; 6,208,990; 6,339,775; 6,640,226; 6,789,096; 6,823,373; 6,850,947; 6,895,471; 7,117,215; 7,162,643; 7,243,110; 7,254,590; 7,281,001; 7,421,458; 7,496,588; 7,523,121; 7,584,422; 7,676,516; 7,720,842; 7,721,270; 7,774,791; 8,065,266; 8,150,803; 8,166,048; 8,166,071; 8,200,622; 8,224,873; 8,271,477; 8,327,419; 8,386,435; 8,392,460; 8,453,159; 8,458,230; 8,707,336; 8,886,617 and RE44,478, International Patents and other Patents Pending.

DISCLAIMER: Informatica Corporation provides this documentation "as is" without warranty of any kind, either express or implied, including, but not limited to, the implied warranties of noninfringement, merchantability, or use for a particular purpose. Informatica Corporation does not warrant that this software or documentation is error free. The information provided in this software or documentation may include technical inaccuracies or typographical errors. The information in this software and documentation is subject to change at any time without notice.

NOTICES

This Informatica product (the "Software") includes certain drivers (the "DataDirect Drivers") from DataDirect Technologies, an operating company of Progress Software Corporation ("DataDirect") which are subject to the following terms and conditions:

1. THE DATADIRECT DRIVERS ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT.
2. IN NO EVENT WILL DATADIRECT OR ITS THIRD PARTY SUPPLIERS BE LIABLE TO THE END-USER CUSTOMER FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL OR OTHER DAMAGES ARISING OUT OF THE USE OF THE ODBC DRIVERS, WHETHER OR NOT INFORMED OF THE POSSIBILITIES OF DAMAGES IN ADVANCE. THESE LIMITATIONS APPLY TO ALL CAUSES OF ACTION, INCLUDING, WITHOUT LIMITATION, BREACH OF CONTRACT, BREACH OF WARRANTY, NEGLIGENCE, STRICT LIABILITY, MISREPRESENTATION AND OTHER TORTS.

Publication Date: 2018-07-28

# Table of Contents


Preface. . . . . 5
Informatica Resources. . . . . 5
Informatica My Support Portal. . . . . 5
Informatica Documentation. . . . . 5
Informatica Product Availability Matrixes. . . . . 5
Informatica Web Site. . . . . 5
Informatica How-To Library. . . . . 5
Informatica Knowledge Base. . . . . 6
Informatica Support YouTube Channel. . . . . 6
Informatica Marketplace. . . . . 6
Informatica Velocity. . . . . 6
Informatica Global Customer Support. . . . . 6

**Chapter 1: Introduction to RulePoint. . . . . 7**

RulePoint Overview. . . . . 7
How RulePoint Works. . . . . 7
RulePoint Example. . . . . 8
RulePoint Architecture. . . . . 8
Design-time Environment. . . . . 8
Run-time Environment. . . . . 8
RulePoint User Interface. . . . . 10
Design Tab. . . . . 10
Administration Tab. . . . . 11
Dashboard Tab. . . . . 12

**Chapter 2: RulePoint Objects. . . . . 13**

RulePoint Objects Overview. . . . . 13
High-Level Workflow for Getting Started with RulePoint. . . . . 15
Sample References. . . . . 17

# Preface

The *RulePoint Getting Started Guide* is written for RulePoint users. It provides a tutorial to help first-time users to familiarize with various RulePoint tasks. The guide assumes that you are familiar with application server and relational database concepts.

# Informatica Resources

## Informatica My Support Portal

As an Informatica customer, you can access the Informatica My Support Portal at http://mysupport.informatica.com.

The site contains product information, user group information, newsletters, access to the Informatica customer support case management system (ATLAS), the Informatica How-To Library, the Informatica Knowledge Base, Informatica Product Documentation, and access to the Informatica user community.

## Informatica Documentation

The Informatica Documentation team makes every effort to create accurate, usable documentation. If you have questions, comments, or ideas about this documentation, contact the Informatica Documentation team through email at infa_documentation@informatica.com. We will use your feedback to improve our documentation. Let us know if we can contact you regarding your comments.

The Documentation team updates documentation as needed. To get the latest documentation for your product, navigate to Product Documentation from http://mysupport.informatica.com.

## Informatica Product Availability Matrixes

Product Availability Matrixes (PAMs) indicate the versions of operating systems, databases, and other types of data sources and targets that a product release supports. You can access the PAMs on the Informatica My Support Portal at https://mysupport.informatica.com/community/my-support/product-availability-matrices.

## Informatica Web Site

You can access the Informatica corporate web site at http://www.informatica.com. The site contains information about Informatica, its background, upcoming events, and sales offices. You will also find product and partner information. The services area of the site includes important information about technical support, training and education, and implementation services.

## Informatica How-To Library

As an Informatica customer, you can access the Informatica How-To Library at http://mysupport.informatica.com. The How-To Library is a collection of resources to help you learn more about Informatica products and features. It includes articles and interactive demonstrations that provide

solutions to common problems, compare features and behaviors, and guide you through performing specific real-world tasks.

## Informatica Knowledge Base

As an Informatica customer, you can access the Informatica Knowledge Base at http://mysupport.informatica.com. Use the Knowledge Base to search for documented solutions to known technical issues about Informatica products. You can also find answers to frequently asked questions, technical white papers, and technical tips. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team through email at KB_Feedback@informatica.com.

## Informatica Support YouTube Channel

You can access the Informatica Support YouTube channel at http://www.youtube.com/user/INFASupport. The Informatica Support YouTube channel includes videos about solutions that guide you through performing specific tasks. If you have questions, comments, or ideas about the Informatica Support YouTube channel, contact the Support YouTube team through email at supportvideos@informatica.com or send a tweet to @INFASupport.

## Informatica Marketplace

The Informatica Marketplace is a forum where developers and partners can share solutions that augment, extend, or enhance data integration implementations. By leveraging any of the hundreds of solutions available on the Marketplace, you can improve your productivity and speed up time to implementation on your projects. You can access Informatica Marketplace at http://www.informaticamarketplace.com.

## Informatica Velocity

You can access Informatica Velocity at http://mysupport.informatica.com. Developed from the real-world experience of hundreds of data management projects, Informatica Velocity represents the collective knowledge of our consultants who have worked with organizations from around the world to plan, develop, deploy, and maintain successful data management solutions. If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

## Informatica Global Customer Support

You can contact a Customer Support Center by telephone or through the Online Support.

Online Support requires a user name and password. You can request a user name and password at http://mysupport.informatica.com.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at http://www.informatica.com/us/services-and-training/support-services/global-support-centers/.

CHAPTER 1

# Introduction to RulePoint

This chapter includes the following topics:


- RulePoint Overview, 7
- RulePoint Architecture, 8
- RulePoint User Interface, 10


## RulePoint Overview

RulePoint is a complex event processing (CEP) solution that delivers real-time operational business intelligence within your enterprise. Business analysts can use RulePoint to maximize the efficiency of their business processes, determine hidden patterns in data, and correlate real-time events that have a negative impact on business.

### How RulePoint Works

RulePoint uses services to collect information from streams of data flowing from different data sources. Each unit of information is considered an event. RulePoint uses the business logic in rules to detect specific events and respond to a user with specific information.

RulePoint consists of a design-time environment and a run-time environment. RulePoint design-time environment comprises administrator tools and the infrastructure that is required to author objects, configure and administer the run-time environment, and deploy objects from the design-time environment to the run-time environment. The run-time environment provides the execution infrastructure that you need to deploy and run rule processing objects.

You can scale the run-time environment across multiple nodes based on the volume and complexity of data that you want to process. You can configure a primary and multiple backup run-time components to deliver high availability to detect events, process rules, and dispatch alerts. If the primary run-time component fails, the backup run-time component continues to process data.

You can also partition rules, sources, and responders across multiple nodes for efficient processing. For example, you can deploy heavy load sources, custom services that are not thread safe, and rules associated with high throughput and risk on select high-availability nodes.

RulePoint uses the asynchronous messaging model that is built into the Ultra Messaging Persistence (UMP) product to move data between nodes.

## RulePoint Example

You can use RulePoint in financial applications to detect events and monitor financial tasks.

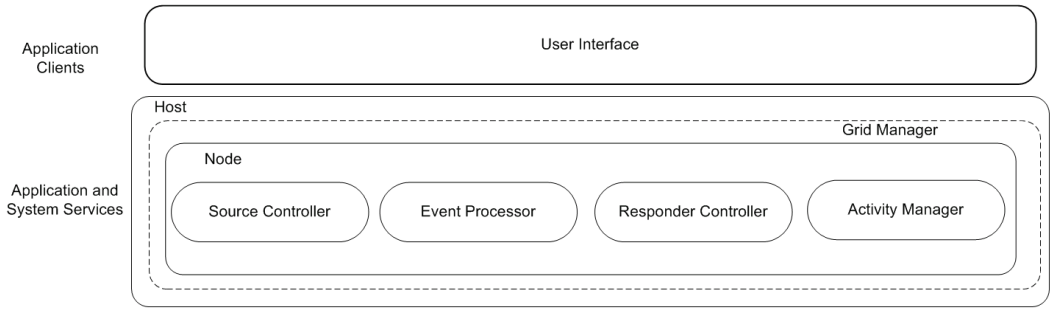
You can configure rules to track streams of data for customer withdrawals that are beyond the required minimum balance and alert the customer immediately. You can also configure rules to block the account and simultaneously alert the customer when the account is accessed with incorrect passwords.

RulePoint provides extensive alerting capabilities through configured channels, such as email, instant messages, or a web browser to individual users, groups, or external applications. Alerts allow the customer to immediately respond and take action.

# RulePoint Architecture

RulePoint consists of a distinct design-time environment and a run-time environment.

The following figure shows the RulePoint design-time and run-time environment in a single node default topology:




## Design-time Environment

The RulePoint design-time environment comprises developer and administrator tools. You can use the design-time user interface to create objects, configure and administer the run-time environment, and deploy objects from the design-time environment to the run-time environment. The design-time environment is packaged as a web application and runs in an Apache Tomcat web container.

You can create primary and secondary objects. Primary objects include sources, responders, and rules. Supporting objects include topics, connections, responses, watchlists, and analytics.

## Run-time Environment

The run-time environment consists of the topology, which provides the execution infrastructure for deploying and processing rules and related objects.

The topology is a grouping of all the application services, system services, nodes, and hosts, along with the entire configuration of RulePoint in the run-time environment. The installer creates a single node topology by default. A single node topology consists of a host, node, application and system services. If you want to scale the topology, you can configure additional hosts, nodes, application services, and system services in a topology.

### Single Node Topology

A single node default topology consists of the following components:

**Host**

A machine where you install RulePoint. The default topology in RulePoint consists of the node and service configurations on the host.

**Node**

The node is embedded in a grid manager and hosts the application services.

**Application Services**

A service that runs on a node in the run-time environment. The RulePoint topology consists of application services that generate and process events, evaluate events against rules, and record system activity.

Application services include the following services:

- Source controller. Generates events.
- Responder controller. Consumes event activations.
- Event processor. Consumes events, runs rules or template rules, and then generates event activations.
- Activity manager. Records and stores system activity for use in reporting.

**System Services**

A service that runs on the host in the run-time environment. System services include the following service:

**Grid Manager**

Controls the life cycle and failover of all application services in the topology. The grid manager manages deployment of objects into the application services and handles all interactions between the design environment and the services.

## Multinode Topology

A multinode default topology can consist of additional hosts, nodes, application and system services. To add additional hosts, first install RulePoint on a host and then copy the RulePoint installation directory to the subsequent hosts. You must then configure the host and additional nodes and services from the topology configurations in the RulePoint Console. The application services can include additional source controllers, responder controllers, activity managers, and event processors. System services include the grid manager and the messaging layer.

Additionally, a multinode topology contains the following contents:

**Host Agent**

A host agent runs on all the hosts in a topology and manages communication between the grid manager and the node. A host agent collects statistics from the processes that run on its host and delivers the statistics to the grid manager. The grid manager sends messages to the host agent based on the statistics it receives. For example, the grid manager might send a message to kill or fork a process on the host.

**Messaging Layer**

The messaging layer helps in communication between the application services. The following system services constitute the messaging layer:

- UM store. Enables guaranteed delivery for data exchange across all services within RulePoint.
- UM lbmrd. Provides resolves address for data exchange across all services within RulePoint.

# RulePoint User Interface

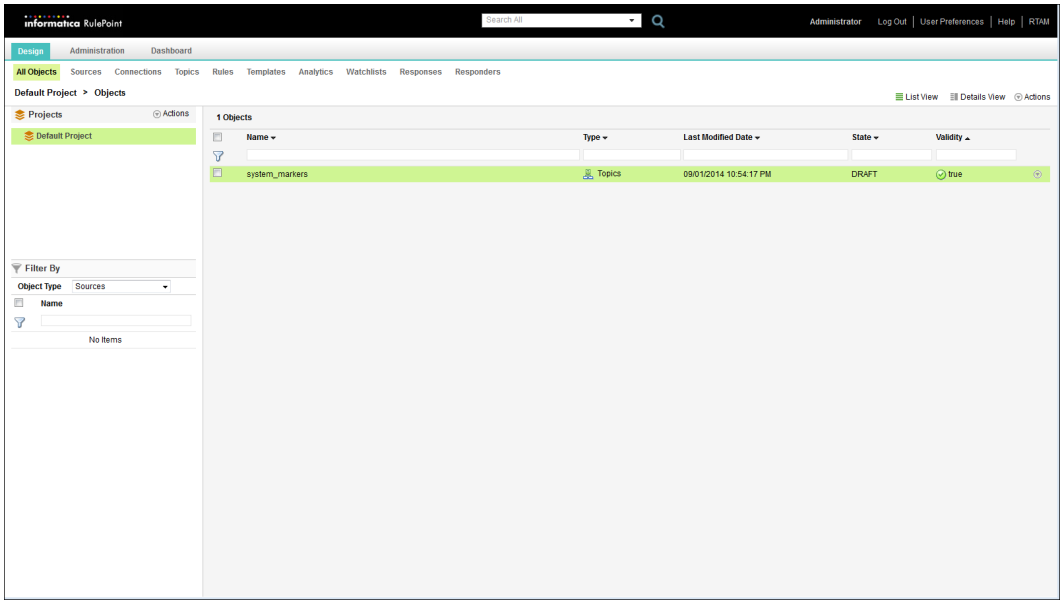
You can use the RulePoint user interface to create users and roles, create and deploy objects, and administer RulePoint.

## Design Tab

Use the **Design** tab to create objects and deploy them to the application services.

The **Design** tab contains views that correspond to a RulePoint object. A RulePoint installation includes a default project and a default topic.

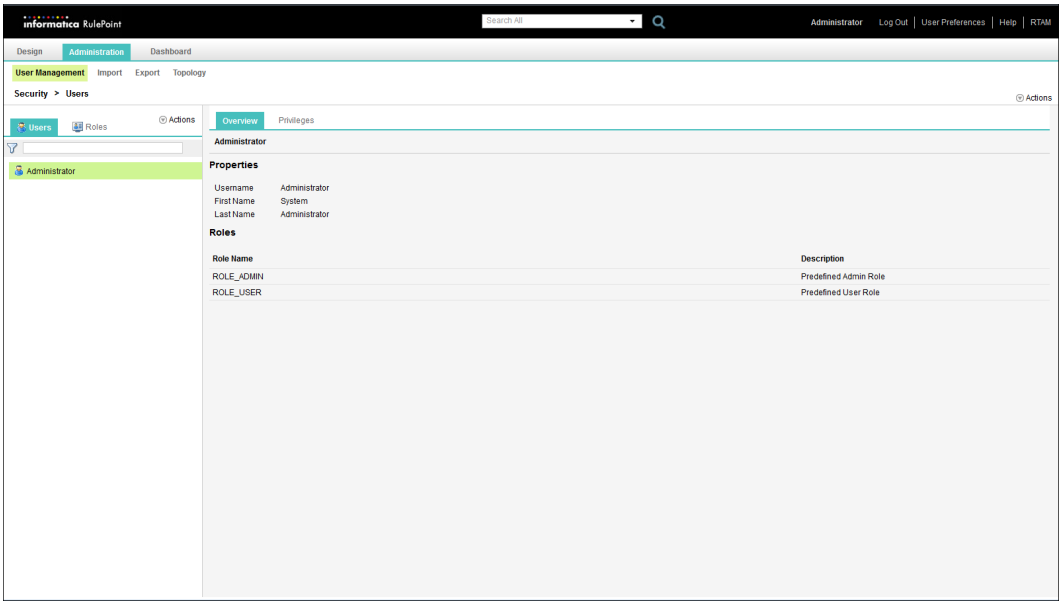
The following figure shows the **All Objects** view that lists the objects in a project:



# Administration Tab

Perform administrative tasks such as manage users, import and export objects, and manage a topology on the **Administration** tab.

Use the **User Management** view, which is available on the **Administration** tab, to view default users and roles. The following figure shows the **User Management** view, which lists the users created in RulePoint:

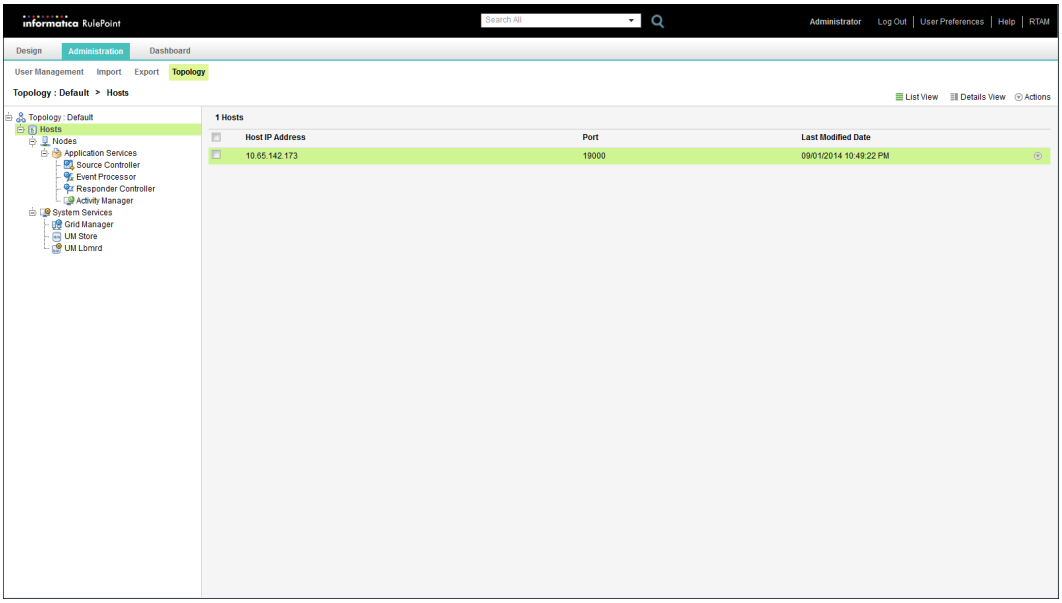



## Topology View

Use the **Topology** view to configure the RulePoint topology.

The RulePoint installer prompts you to create a default topology with a name of your choice. The default topology contains the configuration details of the default host, node, application services, and system services. To view the topology details, click the **Topology** view on the **Administrator** tab.

The following image shows the **Topology** view:



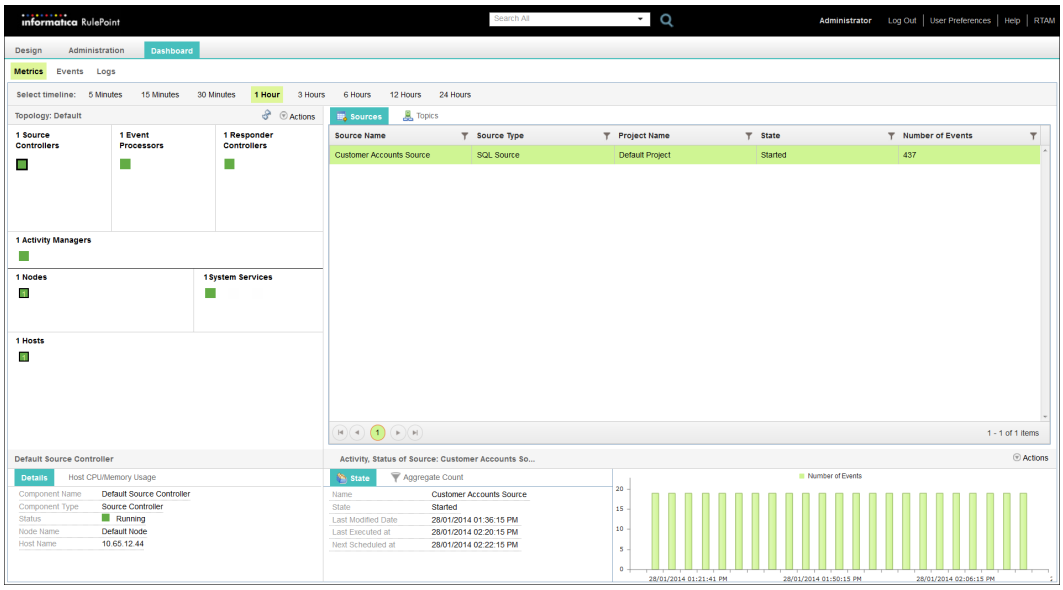
# Dashboard Tab

View application services, system services, nodes, and hosts in the topology on the **Dashboard** tab.

The components appear in green (running), or red (error) states. Pause on a component to view the name of the component, type, and the status. When you select an application service, the nodes and host on which the application service is located appears highlighted. The lower left panel displays the component details, and the CPU and memory utilization for each configured host.

You can view the metrics of objects based on a time line on the right panel. When you select an application service on the left panel, the right panel displays the deployed primary and supporting objects for that application service. The lower right panel displays the activity, status, and a graph of the metrics of a deployed object. You can also administer functions available on the lower-right pane for each object type.

The following figure shows the **Dashboard** tab:



CHAPTER 2

# RulePoint Objects

This chapter includes the following topics:


- RulePoint Objects Overview, 13
- High-Level Workflow for Getting Started with RulePoint, 15
- Sample References, 17


## RulePoint Objects Overview

RulePoint monitors events from sources and delivers responses when events match the rules. You configure a function in RulePoint by creating a specific type of object that suits your requirements.

For example, you create a source to generate a specific set of events that interest you. You create topics to group related events, and you create rules to evaluate those events. You also create responses to those events, and you create responders to deliver the responses. After creating objects, you deploy them to the run-time environment.

RulePoint objects are categorized as primary objects and supporting objects. Sources, responders, and rules are primary objects. Topics, watchlists, analytics, templates (for template rule), connections, and responses are supporting objects.

Primary objects drive the execution of the run-time components. Sources and responders are deployed in their respective service controllers. Rules are deployed in the event processor.

The following table maps a primary object to the corresponding supporting objects:

| Primary Objects | Supporting Objects |
|---|---|
| Source | Topic and Connection |
| Responder | Response and Connection |
| Rule | Topic, Analytic, Watchlist, Response, and Template (for template rule) |

The following table describes the primary and secondary RulePoint objects:

| Object | Description |
|---|---|
| Topics | Topics categorize events and event properties, and provide a logical grouping for events originating from multiple sources. When RulePoint receives an event, it identifies the properties and values of the event and categorizes the event under a specific topic. |
| Connections | Connections connect RulePoint services to the target database. Multiple RulePoint service types can share a single connection. You need to first create a connection and then map that connection to various source objects. |
| Sources | Sources connect to external systems to fetch data, and then convert that data to events. The events are published in topics. Before you create a source, create a topic and a connection for the source.<br>After you create a source, create a schedule to specify intervals at which the source must poll the external sources for data that RulePoint can publish as events. You can configure the following types of schedules:<br>- Configure a dynamic schedule if you want the source to wait for a configured period after completing the previous task and before it begins publishing events again.<br>- Configure a static schedule if you want the source to publish events on a schedule that you set, such as once every 30 minutes. |
| Responder | Responders dispatch alerts to external systems. When an event matches a rule condition, the event processor creates a response, and the responder sends the response to the specified external system.<br>For a responder service and a particular type of response, you define the interface parameters. That is, you specify the action that the responder must take when a rule is activated. From a single responder service, you can create multiple specific responses. |
| Response | After you create a responder service, you must create a response. When a rule invokes that response, the response implements the service with specific field values. |
| Watchlist | A watchlist is a container that stores a set of values as a single object to which you can assign a unique name. You can reference the name in a rule so that the rule can use the data stored in the object. You can modify the values in a watchlist at any time. Any rules that reference that watchlist use the new values. |
| Analytic | An analytic implements a data processing function. Analytics provide simple functions, such as math, text, and date functions, or more advanced functions, such as connecting to a remote server and processing data before returning.<br>You can reference an analytic in a rule to check for various conditions, or manipulate the event data to check specific conditions. |
| Template | If you have experienced with rule syntax, create templates using DRQL. A template includes a rule statement that contains substitution parameters and instructional text to define those parameters. After you create a template, you can easily create new rules using the templates. |
| Rule | You configure rules to identify specific events and respond to those events as they occur. For example, you might want to configure a rule to listen to a news feed and send a specified user an email when the headline of a news story contains "London."<br>You can either use the wizard, template, or advanced rule to create a rule. Configure the following rules based on your requirement:<br>- Use the wizard rule, which is an easy-to-use interface that walks you through the process of selecting topics, conditions, and responses while creating a rule.<br>- Use the advanced rule to create powerful rules that use the Detect and Respond Query Language (DRQL) of RulePoint.<br>- Use the template rule if you are an inexperienced rule writer and want to create rules with minimal errors. You can use the template rule for common scenarios, such as news event alerts, stock quote price checks, and credit card transaction notifications. |

For more information about configuring RulePoint objects and the predefined objects available in RulePoint, see the *RulePoint User Guide*.

# High-Level Workflow for Getting Started with RulePoint

Use RulePoint to create, deploy, import, and export objects, configure the topology components, administer RulePoint, and monitor the workflow progress.

Complete the following tasks to set up the RulePoint design-time and run-time environments:

## Prerequisites

1. Install RulePoint.
2. Make sure that the license file is available in the following directory:

   ```
   <RulePoint installation directory>\conf\
   ```
3. Start the RulePoint server. To start the RulePoint server, you must start the host agent, topology and the design-time instance, in that order. Based on the operating system that you use, perform the following tasks:
   - On Linux, go to the `<RulePoint installation directory>/bin/` directory and perform the following steps:
     1. Enter the following command to start the RulePoint host-agent instance:

        ```
        startHostAgent.sh -h <ip_address> -p <port_number>
        ```
     2. Enter the following command to start the run-time components:

        ```
        topology.sh start <TOPOLOGY_NAME>
        ```
     3. Enter the following command to start the design-time environment:

        ```
        design.sh run
        ```
   - On Windows, go to the `<RulePoint installation directory>\bin\` directory and perform the following steps:
     1. Enter the following command to start the RulePoint host-agent instance:

        ```
        startHostAgent.bat
        ```
     2. Enter the following command to start the run-time components:

        ```
        topology.bat start <TOPOLOGY_NAME>
        ```
     3. Enter the following command to start the design-time environment:

        ```
        design.bat start
        ```

The default topology, which consists of a single-node environment, is up and running.

## Procedure

1. To log in to the RulePoint server, type the following URL in the address bar of your browser: `http://<FQDN:port number>/rulepoint`. Log in to RulePoint by using the default credentials. The default user name is Administrator. The default password is Administrator1.
2. (Optional) Create a role with specific permissions, create a user, and then associate a user with that role. You can log in to RulePoint with those user credentials and create objects.
3. Create a project.

4. Perform the following steps to create primary and secondary objects for the project:
    a. Create a topic to group a particular event, and add the properties that you want to associate with the topic.
    b. Create a connection with properties suited to the source or responder types that you want to create.
    c. Create a RulePoint source to fetch data from an external source.
    d. Create a static or dynamic schedule to specify the interval at which the RulePoint source must poll the external data source for data that can be published as events.
    e. Create a responder to dispatch alerts to external systems.
    f. Create a response, and associate the response with the responder service.
    g. Create rules by using either the wizard rule, template rule, or the advanced rule. You can create the following objects to reference them in rules:
        - (Optional) An analytic, to include math, text, and date functions, or other advanced functions.
        - (Optional) A watchlist, to include a list of values to monitor.
5. Deploy the source, rule, and responder from the design-time environment to the source controller, the event processor, and the responder controller, respectively. You can deploy a single object or multiple objects at a time.

    **Note:** You can redeploy, or undeploy deployed objects.
6. (Optional) You can export or import an object as an XML file, to and from a project, respectively. If you import an XML file, RulePoint uses the XML file to process events and generate alerts.
7. (Optional) If you want to create a multi-node topology and set up a high-availability environment for the application services, perform the following tasks:
    a. Create additional hosts in the topology.

        **Note:** Before you configure a new host in the topology, shut down the topology and design time. Copy the RulePoint installation folder to the host that you want to configure. Set the RulePoint environmental variable *RULEPOINT_HOME* on the host to point to the installation folder and start the host agent on the host.
    b. To start the host agent, navigate to the `<RulePoint installation directory>/bin/` directory, and then run the following command:
        On Windows: `startHostAgent.bat -h <ip_address> -p <port-number>`

        On Linux: `startHostAgent.sh -h <host-name> -p <port-number>`
    c. Create additional nodes.
    d. Select each of the run-time components and set the deployment mode to high availability.
    e. Assign the run-time components to specific nodes based on your requirements.
    f. Based on how you want to distribute the objects across the instances, you can reassign deployed rules, sources, and responders to the event processor and service controller instances, respectively.

        **Note:** You must shut down the topology before you start the configuration. After you complete configuring the components, you must start the topology so that the configurations apply.

        For more information about configuring high availability in RulePoint, see the *RulePoint Administrator Guide*.
8. (Optional) View the dashboard to see how the objects are processed at run time. Use the functions on the dashboard to manage the objects.

    **Note:** You can view the configured application services on the dashboard only when the run time is up and running.

# Sample References

The use cases help you understand the workflow for processing RulePoint objects. The samples help you understand how to use RulePoint to retrieve information from a source, process the information, and notify you with the appropriate alerts.

For more information, see the following documents:

***Banking Use Case Tutorial***

Represents the RulePoint processing flow using data that simulates a banking database and provides solutions for banking requirements. The tutorial helps you understand the RulePoint functionalities. You learn how to create RulePoint objects, deploy the objects, and receive appropriate alerts. You learn to configure custom services in RulePoint and use the REST APIs taking the banking use case as an example.

***Trading Use Case Example***

Presents trading use cases that you can solve using RulePoint.

***Business Process Management Example***

Presents common loan processing scenarios solved using RulePoint.